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Experiment 02

Aim

To containerize the application using Docker.

Learning Outcome

LO4: Containerize, orchestrate, and deploy applications using Docker, Docker Compose, Kubernetes, and cloud infrastructure tools.

Theory

Docker

Docker is an open-source containerization platform that enables applications to run inside isolated environments called containers. Each container packages the application along with its libraries, dependencies, and configuration files, ensuring consistent execution across different systems. Since containers share the host operating system's kernel, they are lightweight, portable, and start much faster than traditional virtual machines.

Docker Image

A Docker image is a read-only template used to create Docker containers. It contains the operating system, application code, libraries, dependencies, configuration files, and runtime environment required to run an application. Images are stored in repositories such as Docker Hub and can be downloaded using the docker pull command. One image can be used to create multiple containers.

Containerization

Containerization is a virtualization technology that packages an application along with all its dependencies into an isolated unit called a container. Unlike virtual machines, containers share the host operating system's kernel, making them lightweight, resource-efficient, portable, and fast to start. Docker is one of the most widely used platforms for containerization and plays an important role in modern DevOps workflows.

Difference between Docker Image and Docker Container

Docker Image	**Docker Container**
Read-only template	Running instance of an image
Contains application files and dependencies	Contains the executing application
Cannot run by itself	Executes the application
Created using a Dockerfile	Created using docker run

Stored in Docker Hub or local repository	Exists while running on the host
One image can create multiple containers	Each container is an independent instance
Read-only	Read-write during execution

Commands Used

Check Docker Version

```
docker --version
```

List Docker Images

```
docker images
```

Download Ubuntu Image

```
docker pull ubuntu
```

Run Ubuntu Container

```
docker run -it ubuntu
```

Linux Commands Inside Container

```
pwd
ls
cd home
mkdir T21_203
cd T21_203
touch roll_no_203.txt
echo "This is my text" > roll_no_203.txt
cat roll_no_203.txt
exit
```

Docker Container Commands

```
docker ps
docker ps -a
docker start unruffled_brown
docker inspect unruffled_brown
docker rm -f unruffled_brown
```

Docker Image and Network Commands

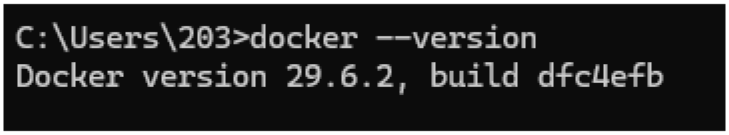
```
docker image prune
docker container ls -a
docker info
docker history ubuntu
docker network inspect bridge
docker network create my_network
```

Observations

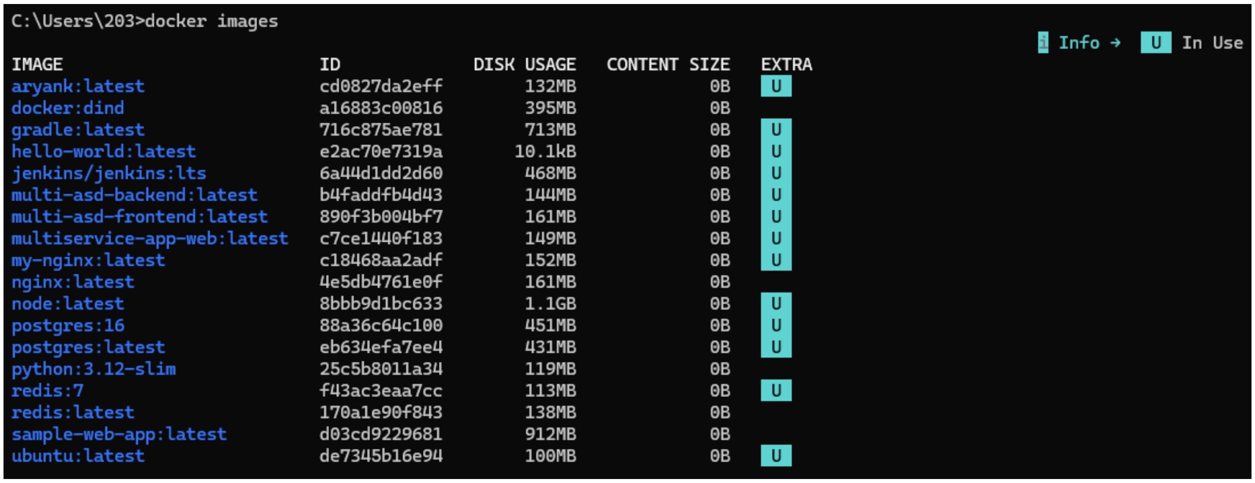
- Docker was successfully installed and verified.
- Ubuntu image was downloaded from Docker Hub.
- An Ubuntu container was created and accessed in interactive mode.
- Basic Linux commands such as pwd, ls, mkdir, touch, echo, and cat were executed successfully inside the container.
- Docker commands were used to inspect, start, and remove containers.
- Docker network information and image history were viewed successfully.

Result

The application environment was successfully containerized using Docker. Docker images and containers were managed using various Docker commands, and basic Linux file operations were performed inside the Ubuntu container. The experiment demonstrated the concepts of Docker images, containers, and containerization.



```
C:\Users\203>docker --version
Docker version 29.6.2, build dfc4efb
```



```
C:\Users\203>docker images
                                                                                    i Info →  U  In Use
IMAGE                         ID             DISK USAGE   CONTENT SIZE   EXTRA
aryank:latest                 cd0827da2eff       132MB             0B    U
docker:dind                   a16883c00816       395MB             0B
gradle:latest                 716c875ae781       713MB             0B    U
hello-world:latest            e2ac70e7319a      10.1kB             0B    U
jenkins/jenkins:lts           6a44d1dd2d60       468MB             0B    U
multi-asd-backend:latest      b4faddfb4d43       144MB             0B    U
multi-asd-frontend:latest     890f3b004bf7       161MB             0B    U
multiservice-app-web:latest   c7ce1440f183       149MB             0B    U
my-nginx:latest               c18468aa2adf       152MB             0B    U
nginx:latest                  4e5db4761e0f       161MB             0B
node:latest                   8bbb9d1bc633       1.1GB             0B    U
postgres:16                   88a36c64c100       451MB             0B    U
postgres:latest               eb634efa7ee4       431MB             0B    U
python:3.12-slim              25c5b8011a34       119MB             0B
redis:7                       f43ac3eaa7cc       113MB             0B    U
redis:latest                  170a1e90f843       138MB             0B
sample-web-app:latest         d03cd9229681       912MB             0B
ubuntu:latest                 de7345b16e94       100MB             0B    U
```

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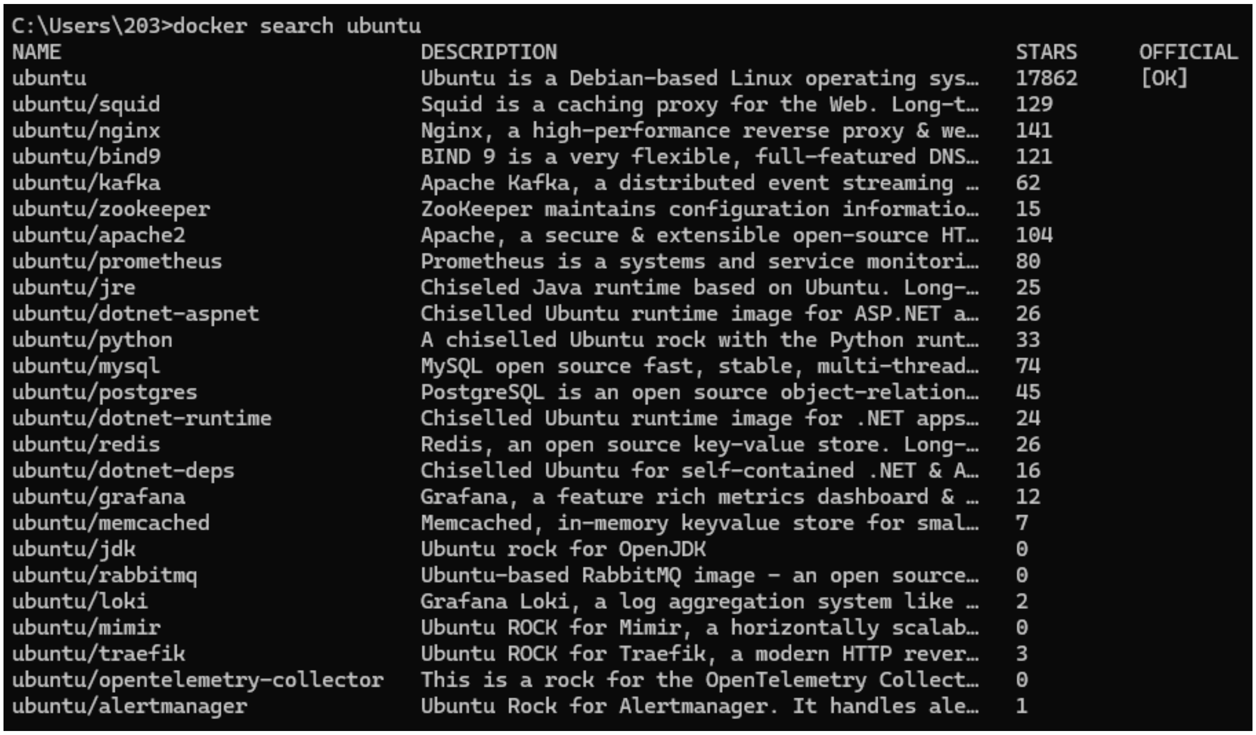
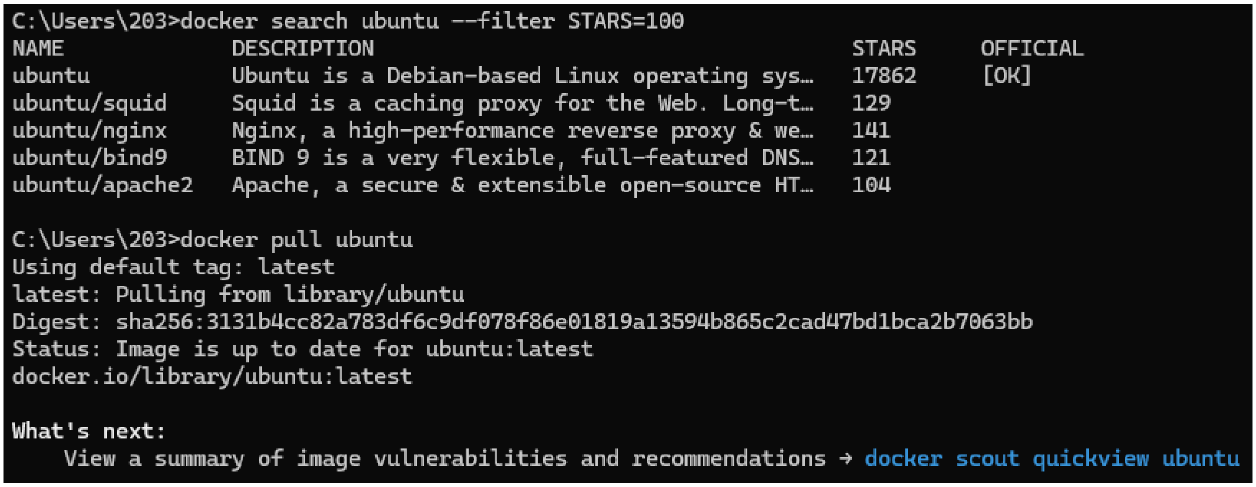


```
C:\Users\203>docker search ubuntu
NAME                             DESCRIPTION                                     STARS     OFFICIAL
ubuntu                           Ubuntu is a Debian-based Linux operating sys…   17862     [OK]
ubuntu/squid                     Squid is a caching proxy for the Web. Long-t…   129
ubuntu/nginx                     Nginx, a high-performance reverse proxy & we…   141
ubuntu/bind9                     BIND 9 is a very flexible, full-featured DNS…   121
ubuntu/kafka                     Apache Kafka, a distributed event streaming …   62
ubuntu/zookeeper                 ZooKeeper maintains configuration informatio…   15
ubuntu/apache2                   Apache, a secure & extensible open-source HT…   104
ubuntu/prometheus                Prometheus is a systems and service monitori…   80
ubuntu/jre                       Chiseled Java runtime based on Ubuntu. Long-…   25
ubuntu/dotnet-aspnet             Chiselled Ubuntu runtime image for ASP.NET a…   26
ubuntu/python                    A chiselled Ubuntu rock with the Python runt…   33
ubuntu/mysql                     MySQL open source fast, stable, multi-thread…   74
ubuntu/postgres                  PostgreSQL is an open source object-relation…   45
ubuntu/dotnet-runtime            Chiselled Ubuntu runtime image for .NET apps…   24
ubuntu/redis                     Redis, an open source key-value store. Long-…   26
ubuntu/dotnet-deps               Chiselled Ubuntu for self-contained .NET & A…   16
ubuntu/grafana                   Grafana, a feature rich metrics dashboard & …   12
ubuntu/memcached                 Memcached, in-memory keyvalue store for smal…   7
ubuntu/jdk                       Ubuntu rock for OpenJDK                         0
ubuntu/rabbitmq                  Ubuntu-based RabbitMQ image - an open source…   0
ubuntu/loki                      Grafana Loki, a log aggregation system like …   2
ubuntu/mimir                     Ubuntu ROCK for Mimir, a horizontally scalab…   0
ubuntu/traefik                   Ubuntu ROCK for Traefik, a modern HTTP rever…   3
ubuntu/opentelemetry-collector   This is a rock for the OpenTelemetry Collect…   0
ubuntu/alertmanager              Ubuntu Rock for Alertmanager. It handles ale…   1
```



```
C:\Users\203>docker search ubuntu --filter STARS=100
NAME             DESCRIPTION                                     STARS     OFFICIAL
ubuntu           Ubuntu is a Debian-based Linux operating sys…   17862     [OK]
ubuntu/squid     Squid is a caching proxy for the Web. Long-t…   129
ubuntu/nginx     Nginx, a high-performance reverse proxy & we…   141
ubuntu/bind9     BIND 9 is a very flexible, full-featured DNS…   121
ubuntu/apache2   Apache, a secure & extensible open-source HT…   104

C:\Users\203>docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
Digest: sha256:3131b4cc82a783df6c9df078f86e01819a13594b865c2cad47bd1bca2b7063bb
Status: Image is up to date for ubuntu:latest
docker.io/library/ubuntu:latest

What's next:
    View a summary of image vulnerabilities and recommendations → docker scout quickview ubuntu
```

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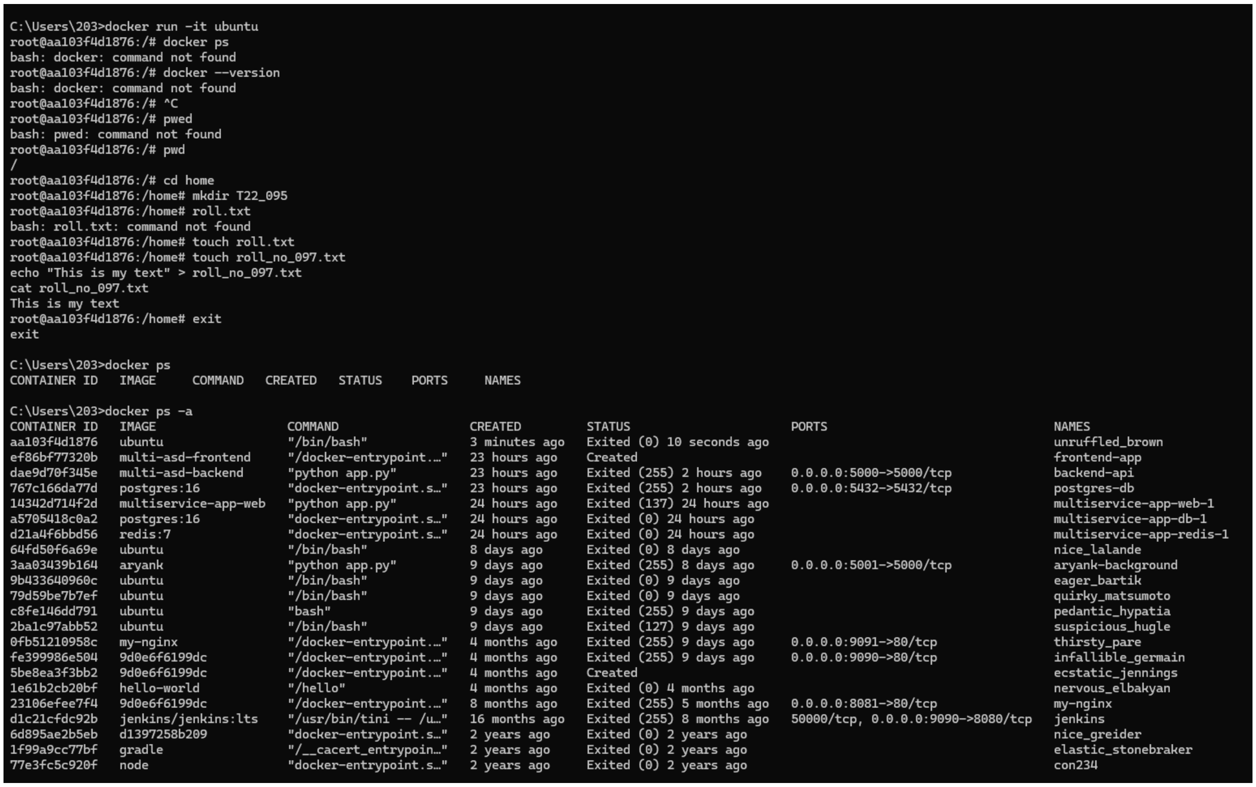
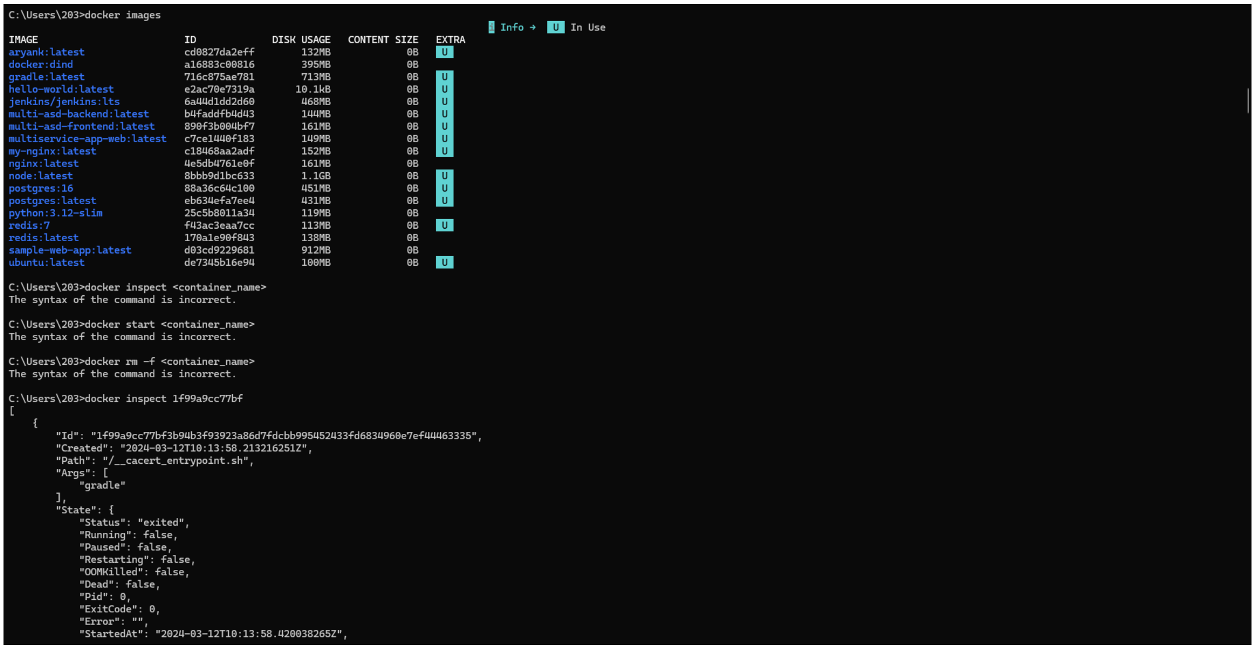


```
C:\Users\203>docker run -it ubuntu
root@aa103f4d1876:/# docker ps
bash: docker: command not found
root@aa103f4d1876:/# docker --version
bash: docker: command not found
root@aa103f4d1876:/# ^C
root@aa103f4d1876:/# pwed
bash: pwed: command not found
root@aa103f4d1876:/# pwd
/
root@aa103f4d1876:/# cd home
root@aa103f4d1876:/home# mkdir T22_095
root@aa103f4d1876:/home# roll.txt
bash: roll.txt: command not found
root@aa103f4d1876:/home# touch roll.txt
root@aa103f4d1876:/home# touch roll_no_097.txt
echo "This is my text" > roll_no_097.txt
cat roll_no_097.txt
This is my text
root@aa103f4d1876:/home# exit
exit

C:\Users\203>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES

C:\Users\203>docker ps -a
CONTAINER ID   IMAGE                  COMMAND                  CREATED          STATUS                      PORTS                               NAMES
aa103f4d1876   ubuntu                 "/bin/bash"              3 minutes ago    Exited (0) 10 seconds ago                                       unruffled_brown
ef86bf77320b   multi-asd-frontend     "/docker-entrypoint.…"   23 hours ago     Created                                                         frontend-app
dae9d70f345e   multi-asd-backend      "python app.py"          23 hours ago     Exited (255) 2 hours ago    0.0.0.0:5000->5000/tcp              backend-api
767c166da77d   postgres:16            "docker-entrypoint.s…"   23 hours ago     Exited (255) 2 hours ago    0.0.0.0:5432->5432/tcp              postgres-db
14342d714f2d   multiservice-app-web   "python app.py"          24 hours ago     Exited (137) 24 hours ago                                       multiservice-app-web-1
a5705418c0a2   postgres:16            "docker-entrypoint.s…"   24 hours ago     Exited (0) 24 hours ago                                         multiservice-app-db-1
d21a4f6bbd56   redis:7                "docker-entrypoint.s…"   24 hours ago     Exited (0) 24 hours ago                                         multiservice-app-redis-1
64fd50f6a69e   ubuntu                 "/bin/bash"              8 days ago       Exited (0) 8 days ago                                           nice_lalande
3aa03439b164   aryank                 "python app.py"          9 days ago       Exited (255) 8 days ago     0.0.0.0:5001->5000/tcp              aryank-background
9b433640960c   ubuntu                 "/bin/bash"              9 days ago       Exited (0) 9 days ago                                           eager_bartik
79d59be7b7ef   ubuntu                 "/bin/bash"              9 days ago       Exited (0) 9 days ago                                           quirky_matsumoto
c8fe146dd791   ubuntu                 "bash"                   9 days ago       Exited (255) 9 days ago                                         pedantic_hypatia
2ba1c97abb52   ubuntu                 "/bin/bash"              9 days ago       Exited (127) 9 days ago                                         suspicious_hugle
0fb51210958c   my-nginx               "/docker-entrypoint.…"   4 months ago     Exited (255) 9 days ago     0.0.0.0:9091->80/tcp                thirsty_pare
fe399986e504   9d0e6f6199dc           "/docker-entrypoint.…"   4 months ago     Exited (255) 9 days ago     0.0.0.0:9090->80/tcp                infallible_germain
5be8ea3f3bb2   9d0e6f6199dc           "/docker-entrypoint.…"   4 months ago     Created                                                         ecstatic_jennings
1e61b2cb20bf   hello-world            "/hello"                 4 months ago     Exited (0) 4 months ago                                         nervous_elbakyan
23106efee7f4   9d0e6f6199dc           "/docker-entrypoint.…"   8 months ago     Exited (255) 5 months ago   0.0.0.0:8081->80/tcp                my-nginx
d1c21cfdc92b   jenkins/jenkins:lts    "/usr/bin/tini -- /u…"   16 months ago    Exited (255) 8 months ago   50000/tcp, 0.0.0.0:9090->8080/tcp   jenkins
6d895ae2b5eb   d1397258b209           "docker-entrypoint.s…"   2 years ago      Exited (0) 2 years ago                                          nice_greider
1f99a9cc77bf   gradle                 "/__cacert_entrypoin…"   2 years ago      Exited (0) 2 years ago                                          elastic_stonebraker
77e3fc5c920f   node                   "docker-entrypoint.s…"   2 years ago      Exited (0) 2 years ago                                          con234
```



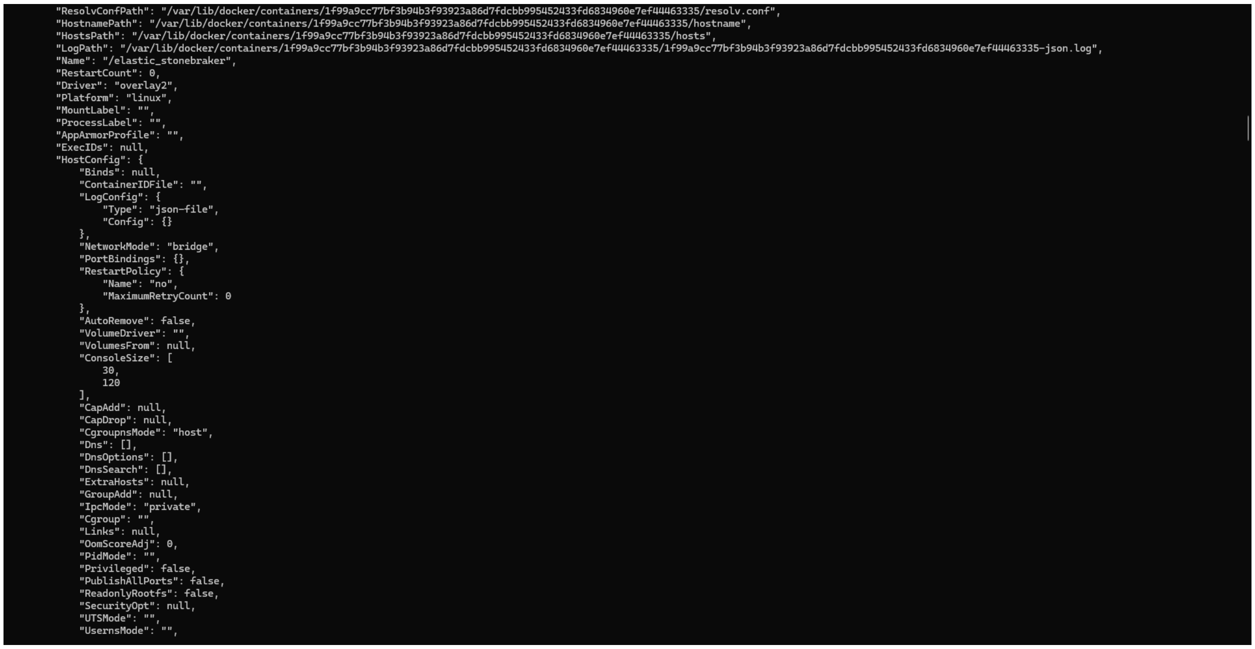
```
C:\Users\203>docker images
                                                                                                   i Info →   U  In Use
IMAGE                         ID             DISK USAGE   CONTENT SIZE   EXTRA
aryank:latest                 cd0827da2eff        132MB             0B    U
docker:dind                   a16883c00816        395MB             0B
gradle:latest                 716c875ae781        713MB             0B    U
hello-world:latest            e2ac70e7319a       10.1kB             0B    U
jenkins/jenkins:lts           6a44d1dd2d60        468MB             0B    U
multi-asd-backend:latest      b4faddfb4d43        144MB             0B    U
multi-asd-frontend:latest     890f3b004bf7        161MB             0B    U
multiservice-app-web:latest   c7ce1440f183        149MB             0B    U
my-nginx:latest               c18468aa2adf        152MB             0B    U
nginx:latest                  4e5db4761e0f        161MB             0B
node:latest                   8bbb9d1bc633        1.1GB             0B    U
postgres:16                   88a36c64c100        451MB             0B    U
postgres:latest               eb634efa7ee4        431MB             0B    U
python:3.12-slim              25c5b8011a34        119MB             0B
redis:7                       f43ac3eaa7cc        113MB             0B    U
redis:latest                  170a1e90f843        138MB             0B
sample-web-app:latest         d03cd9229681        912MB             0B
ubuntu:latest                 de7345b16e94        100MB             0B    U

C:\Users\203>docker inspect <container_name>
The syntax of the command is incorrect.

C:\Users\203>docker start <container_name>
The syntax of the command is incorrect.

C:\Users\203>docker rm -f <container_name>
The syntax of the command is incorrect.

C:\Users\203>docker inspect 1f99a9cc77bf
[
    {
        "Id": "1f99a9cc77bf3b94b3f93923a86d7fdcbb995452433fd6834960e7ef44463335",
        "Created": "2024-03-12T10:13:58.213216251Z",
        "Path": "/__cacert_entrypoint.sh",
        "Args": [
            "gradle"
        ],
        "State": {
            "Status": "exited",
            "Running": false,
            "Paused": false,
            "Restarting": false,
            "OOMKilled": false,
            "Dead": false,
            "Pid": 0,
            "ExitCode": 0,
            "Error": "",
            "StartedAt": "2024-03-12T10:13:58.420038265Z",
```



```
        "ResolvConfPath": "/var/lib/docker/containers/1f99a9cc77bf3b94b3f93923a86d7fdcbb995452433fd6834960e7ef44463335/resolv.conf",
        "HostnamePath": "/var/lib/docker/containers/1f99a9cc77bf3b94b3f93923a86d7fdcbb995452433fd6834960e7ef44463335/hostname",
        "HostsPath": "/var/lib/docker/containers/1f99a9cc77bf3b94b3f93923a86d7fdcbb995452433fd6834960e7ef44463335/hosts",
        "LogPath": "/var/lib/docker/containers/1f99a9cc77bf3b94b3f93923a86d7fdcbb995452433fd6834960e7ef44463335/1f99a9cc77bf3b94b3f93923a86d7fdcbb995452433fd6834960e7ef44463335-json.log",
        "Name": "/elastic_stonebraker",
        "RestartCount": 0,
        "Driver": "overlay2",
        "Platform": "linux",
        "MountLabel": "",
        "ProcessLabel": "",
        "AppArmorProfile": "",
        "ExecIDs": null,
        "HostConfig": {
            "Binds": null,
            "ContainerIDFile": "",
            "LogConfig": {
                "Type": "json-file",
                "Config": {}
            },
            "NetworkMode": "bridge",
            "PortBindings": {},
            "RestartPolicy": {
                "Name": "no",
                "MaximumRetryCount": 0
            },
            "AutoRemove": false,
            "VolumeDriver": "",
            "VolumesFrom": null,
            "ConsoleSize": [
                30,
                120
            ],
            "CapAdd": null,
            "CapDrop": null,
            "CgroupnsMode": "host",
            "Dns": [],
            "DnsOptions": [],
            "DnsSearch": [],
            "ExtraHosts": null,
            "GroupAdd": null,
            "IpcMode": "private",
            "Cgroup": "",
            "Links": null,
            "OomScoreAdj": 0,
            "PidMode": "",
            "Privileged": false,
            "PublishAllPorts": false,
            "ReadonlyRootfs": false,
            "SecurityOpt": null,
            "UTSMode": "",
            "UsernsMode": "",
```

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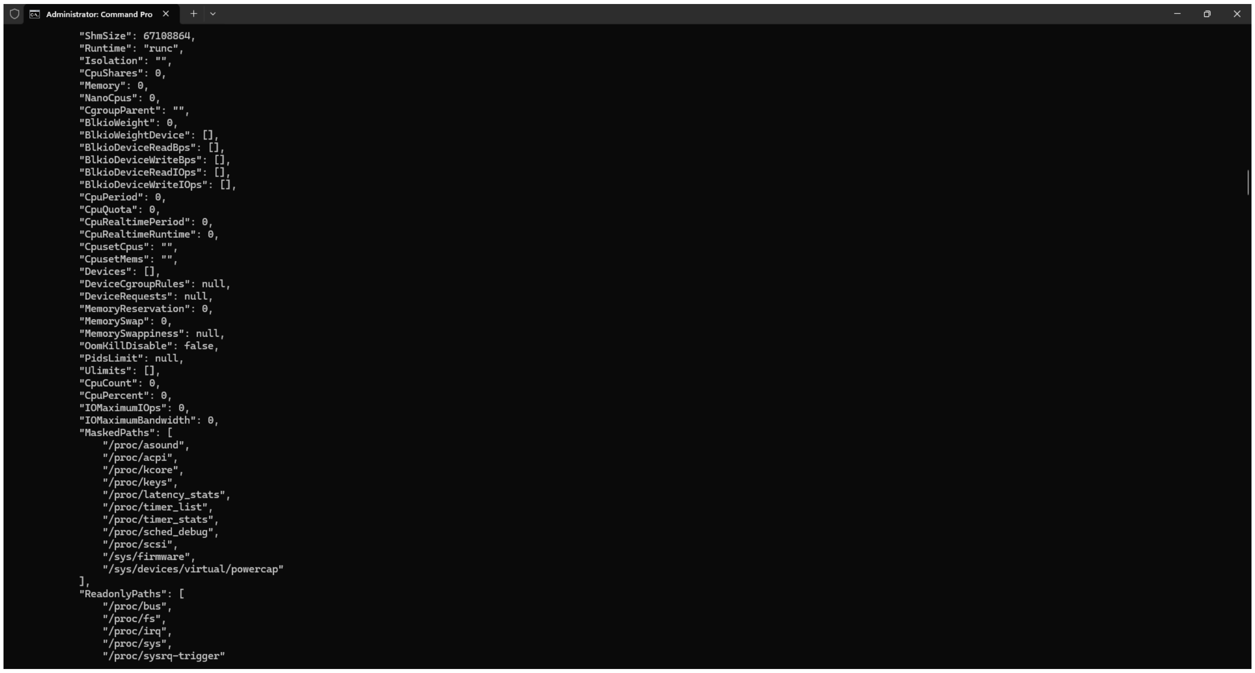
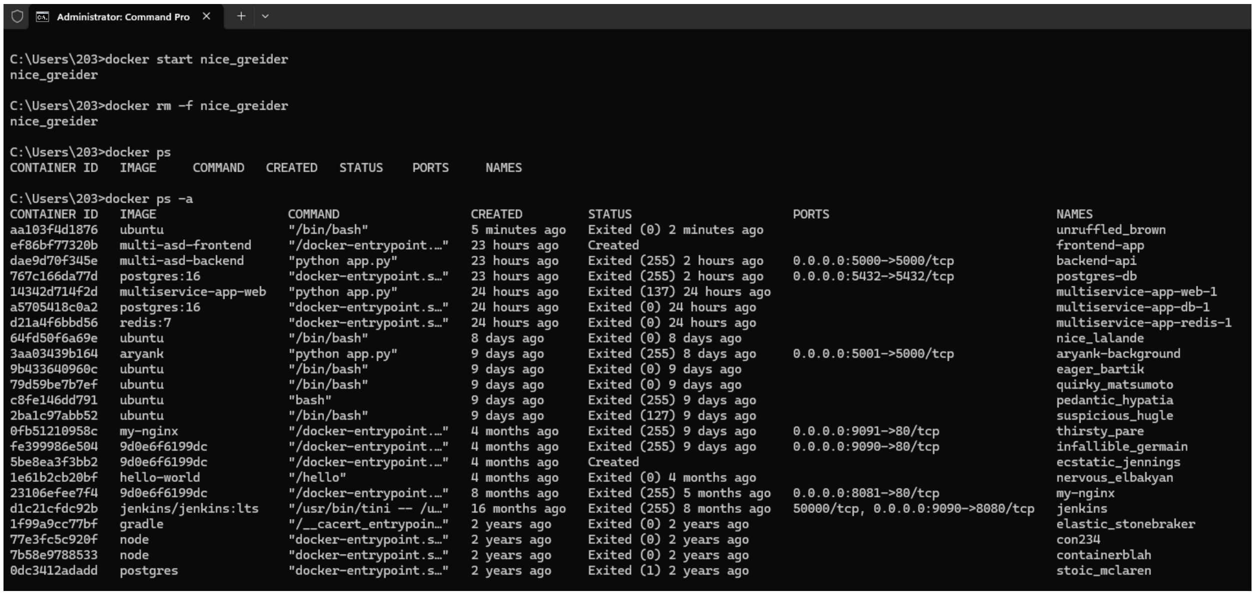
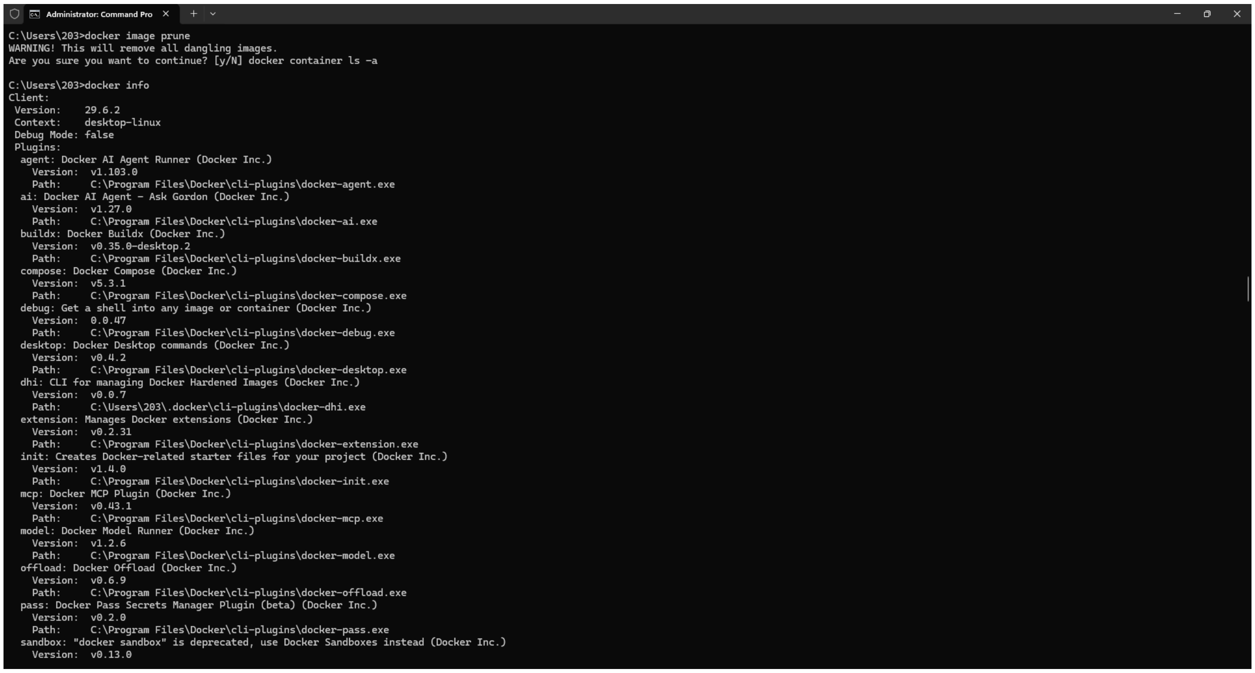


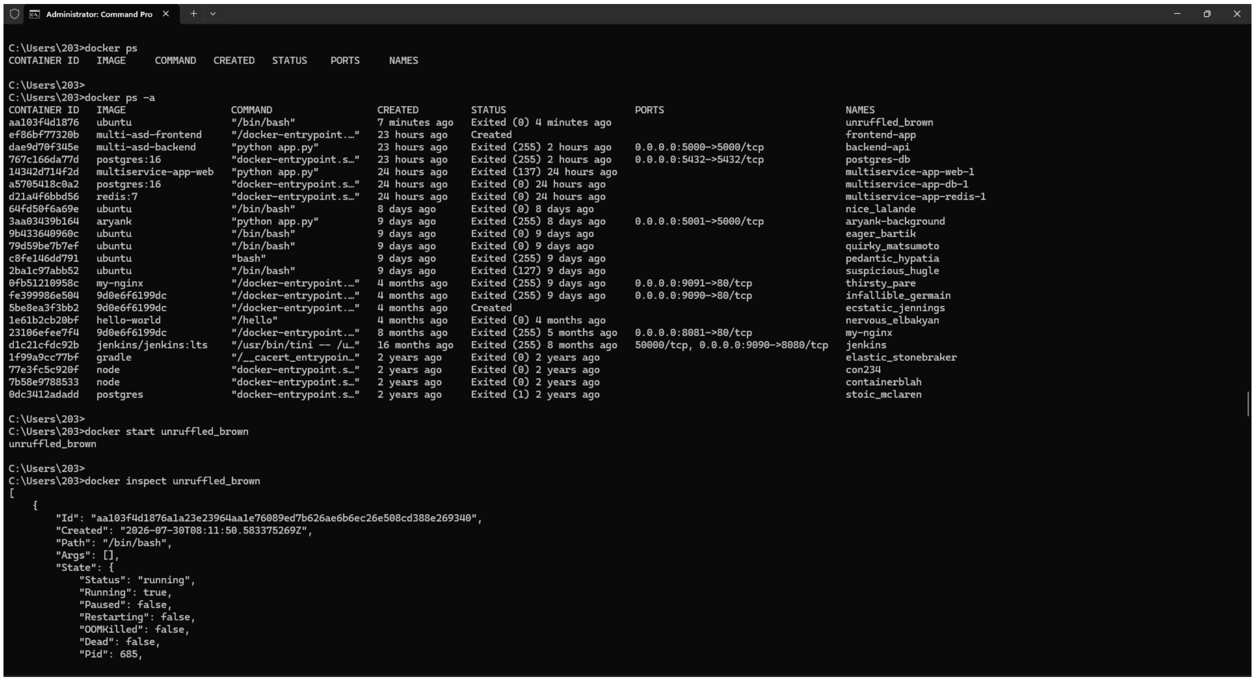
```
        "ShmSize": 67108864,
        "Runtime": "runc",
        "Isolation": "",
        "CpuShares": 0,
        "Memory": 0,
        "NanoCpus": 0,
        "CgroupParent": "",
        "BlkioWeight": 0,
        "BlkioWeightDevice": [],
        "BlkioDeviceReadBps": [],
        "BlkioDeviceWriteBps": [],
        "BlkioDeviceReadIOps": [],
        "BlkioDeviceWriteIOps": [],
        "CpuPeriod": 0,
        "CpuQuota": 0,
        "CpuRealtimePeriod": 0,
        "CpuRealtimeRuntime": 0,
        "CpusetCpus": "",
        "CpusetMems": "",
        "Devices": [],
        "DeviceCgroupRules": null,
        "DeviceRequests": null,
        "MemoryReservation": 0,
        "MemorySwap": 0,
        "MemorySwappiness": null,
        "OomKillDisable": false,
        "PidsLimit": null,
        "Ulimits": [],
        "CpuCount": 0,
        "CpuPercent": 0,
        "IOMaximumIOps": 0,
        "IOMaximumBandwidth": 0,
        "MaskedPaths": [
            "/proc/asound",
            "/proc/acpi",
            "/proc/kcore",
            "/proc/keys",
            "/proc/latency_stats",
            "/proc/timer_list",
            "/proc/timer_stats",
            "/proc/sched_debug",
            "/proc/scsi",
            "/sys/firmware",
            "/sys/devices/virtual/powercap"
        ],
        "ReadonlyPaths": [
            "/proc/bus",
            "/proc/fs",
            "/proc/irq",
            "/proc/sys",
            "/proc/sysrq-trigger"
```



```
C:\Users\203>docker start nice_greider
nice_greider

C:\Users\203>docker rm -f nice_greider
nice_greider

C:\Users\203>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES

C:\Users\203>docker ps -a
CONTAINER ID   IMAGE                  COMMAND                  CREATED         STATUS                       PORTS                               NAMES
aa103f4d1876   ubuntu                 "/bin/bash"              5 minutes ago   Exited (0) 2 minutes ago                                         unruffled_brown
ef86bf77320b   multi-asd-frontend     "/docker-entrypoint.…"   23 hours ago    Created                                                          frontend-app
dae9d70f345e   multi-asd-backend      "python app.py"          23 hours ago    Exited (255) 2 hours ago     0.0.0.0:5000->5000/tcp              backend-api
767c166da77d   postgres:16            "docker-entrypoint.s…"   23 hours ago    Exited (255) 2 hours ago     0.0.0.0:5432->5432/tcp              postgres-db
14342d714f2d   multiservice-app-web   "python app.py"          24 hours ago    Exited (137) 24 hours ago                                        multiservice-app-web-1
a5705418c0a2   postgres:16            "docker-entrypoint.s…"   24 hours ago    Exited (0) 24 hours ago                                          multiservice-app-db-1
d21a4f6bbd56   redis:7                "docker-entrypoint.s…"   24 hours ago    Exited (0) 24 hours ago                                          multiservice-app-redis-1
64fd50f6a69e   ubuntu                 "/bin/bash"              8 days ago      Exited (0) 8 days ago                                            nice_lalande
3aa03439b164   aryank                 "python app.py"          9 days ago      Exited (255) 8 days ago      0.0.0.0:5001->5000/tcp              aryank-background
9b433640960c   ubuntu                 "/bin/bash"              9 days ago      Exited (0) 9 days ago                                            eager_bartik
79d59be7b7ef   ubuntu                 "/bin/bash"              9 days ago      Exited (0) 9 days ago                                            quirky_matsumoto
c8fe146dd791   ubuntu                 "bash"                   9 days ago      Exited (255) 9 days ago                                          pedantic_hypatia
2ba1c97abb52   ubuntu                 "/bin/bash"              9 days ago      Exited (127) 9 days ago                                          suspicious_hugle
0fb51210958c   my-nginx               "/docker-entrypoint.…"   4 months ago    Exited (255) 9 days ago      0.0.0.0:9091->80/tcp                thirsty_pare
fe399986e504   9d0e6f6199dc           "/docker-entrypoint.…"   4 months ago    Exited (255) 9 days ago      0.0.0.0:9090->80/tcp                infallible_germain
5be8ea3f3bb2   9d0e6f6199dc           "/docker-entrypoint.…"   4 months ago    Created                                                          ecstatic_jennings
1e61b2cb20bf   hello-world            "/hello"                 4 months ago    Exited (0) 4 months ago                                          nervous_elbakyan
23106efee7f4   9d0e6f6199dc           "/docker-entrypoint.…"   8 months ago    Exited (255) 5 months ago    0.0.0.0:8081->80/tcp                my-nginx
d1c21cfdc92b   jenkins/jenkins:lts    "/usr/bin/tini -- /u…"   16 months ago   Exited (255) 8 months ago    50000/tcp, 0.0.0.0:9090->8080/tcp   jenkins
1f99a9cc77bf   gradle                 "/__cacert_entrypoin…"   2 years ago     Exited (0) 2 years ago                                           elastic_stonebraker
77e3fc5c920f   node                   "docker-entrypoint.s…"   2 years ago     Exited (0) 2 years ago                                           con234
7b58e9788533   node                   "docker-entrypoint.s…"   2 years ago     Exited (0) 2 years ago                                           containerblah
0dc3412adadd   postgres               "docker-entrypoint.s…"   2 years ago     Exited (1) 2 years ago                                           stoic_mclaren
```



```
C:\Users\203>docker image prune
WARNING! This will remove all dangling images.
Are you sure you want to continue? [y/N] docker container ls -a

C:\Users\203>docker info
Client:
 Version:    29.6.2
 Context:    desktop-linux
 Debug Mode: false
 Plugins:
  agent: Docker AI Agent Runner (Docker Inc.)
    Version:  v1.103.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-agent.exe
  ai: Docker AI Agent - Ask Gordon (Docker Inc.)
    Version:  v1.27.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-ai.exe
  buildx: Docker Buildx (Docker Inc.)
    Version:  v0.35.0-desktop.2
    Path:     C:\Program Files\Docker\cli-plugins\docker-buildx.exe
  compose: Docker Compose (Docker Inc.)
    Version:  v5.3.1
    Path:     C:\Program Files\Docker\cli-plugins\docker-compose.exe
  debug: Get a shell into any image or container (Docker Inc.)
    Version:  0.0.47
    Path:     C:\Program Files\Docker\cli-plugins\docker-debug.exe
  desktop: Docker Desktop commands (Docker Inc.)
    Version:  v0.4.2
    Path:     C:\Program Files\Docker\cli-plugins\docker-desktop.exe
  dhi: CLI for managing Docker Hardened Images (Docker Inc.)
    Version:  v0.0.7
    Path:     C:\Users\203\.docker\cli-plugins\docker-dhi.exe
  extension: Manages Docker extensions (Docker Inc.)
    Version:  v0.2.31
    Path:     C:\Program Files\Docker\cli-plugins\docker-extension.exe
  init: Creates Docker-related starter files for your project (Docker Inc.)
    Version:  v1.4.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-init.exe
  mcp: Docker MCP Plugin (Docker Inc.)
    Version:  v0.43.1
    Path:     C:\Program Files\Docker\cli-plugins\docker-mcp.exe
  model: Docker Model Runner (Docker Inc.)
    Version:  v1.2.6
    Path:     C:\Program Files\Docker\cli-plugins\docker-model.exe
  offload: Docker Offload (Docker Inc.)
    Version:  v0.6.9
    Path:     C:\Program Files\Docker\cli-plugins\docker-offload.exe
  pass: Docker Pass Secrets Manager Plugin (beta) (Docker Inc.)
    Version:  v0.2.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-pass.exe
  sandbox: "docker sandbox" is deprecated, use Docker Sandboxes instead (Docker Inc.)
    Version:  v0.13.0
```



```
C:\Users\203>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES

C:\Users\203>
C:\Users\203>docker ps -a
CONTAINER ID   IMAGE                  COMMAND                  CREATED         STATUS                      PORTS                               NAMES
aa103f4d1876   ubuntu                 "/bin/bash"              7 minutes ago   Exited (0) 4 minutes ago                                        unruffled_brown
ef86bf77320b   multi-asd-frontend     "/docker-entrypoint.…"   23 hours ago    Created                                                         frontend-app
dae9d70f345e   multi-asd-backend      "python app.py"          23 hours ago    Exited (255) 2 hours ago    0.0.0.0:5000->5000/tcp              backend-api
767c166da77d   postgres:16            "docker-entrypoint.s…"   23 hours ago    Exited (255) 2 hours ago    0.0.0.0:5432->5432/tcp              postgres-db
14342d714f2d   multiservice-app-web   "python app.py"          24 hours ago    Exited (137) 24 hours ago                                       multiservice-app-web-1
a5705418c0a2   postgres:16            "docker-entrypoint.s…"   24 hours ago    Exited (0) 24 hours ago                                         multiservice-app-db-1
d21a4f6bbd56   redis:7                "docker-entrypoint.s…"   24 hours ago    Exited (0) 24 hours ago                                         multiservice-app-redis-1
64fd50f6a69e   ubuntu                 "/bin/bash"              8 days ago      Exited (0) 8 days ago                                           nice_lalande
3aa03439b164   aryank                 "python app.py"          9 days ago      Exited (255) 8 days ago     0.0.0.0:5001->5000/tcp              aryank-background
9b433640960c   ubuntu                 "/bin/bash"              9 days ago      Exited (0) 9 days ago                                           eager_bartik
79d59be7b7ef   ubuntu                 "/bin/bash"              9 days ago      Exited (0) 9 days ago                                           quirky_matsumoto
c8fe146dd791   ubuntu                 "bash"                   9 days ago      Exited (255) 9 days ago                                         pedantic_hypatia
2ba1c97abb52   ubuntu                 "/bin/bash"              9 days ago      Exited (127) 9 days ago                                         suspicious_hugle
0fb51210958c   my-nginx               "/docker-entrypoint.…"   4 months ago    Exited (255) 9 days ago     0.0.0.0:9091->80/tcp                thirsty_pare
fe399986e504   9d0e6f6199dc           "/docker-entrypoint.…"   4 months ago    Exited (255) 9 days ago     0.0.0.0:9090->80/tcp                infallible_germain
5be8ea3f3bb2   9d0e6f6199dc           "/docker-entrypoint.…"   4 months ago    Created                                                         ecstatic_jennings
1e61b2cb20bf   hello-world            "/hello"                 4 months ago    Exited (0) 4 months ago                                         nervous_elbakyan
23106efee7f4   9d0e6f6199dc           "/docker-entrypoint.…"   8 months ago    Exited (255) 5 months ago   0.0.0.0:8081->80/tcp                my-nginx
d1c21cfdc92b   jenkins/jenkins:lts    "/usr/bin/tini -- /u…"   16 months ago   Exited (255) 8 months ago   50000/tcp, 0.0.0.0:9090->8080/tcp   jenkins
1f99a9cc77bf   gradle                 "/__cacert_entrypoin…"   2 years ago     Exited (0) 2 years ago                                          elastic_stonebraker
77e3fc5c920f   node                   "docker-entrypoint.s…"   2 years ago     Exited (0) 2 years ago                                          con234
7b58e9788533   node                   "docker-entrypoint.s…"   2 years ago     Exited (0) 2 years ago                                          containerblah
0dc3412adadd   postgres               "docker-entrypoint.s…"   2 years ago     Exited (1) 2 years ago                                          stoic_mclaren

C:\Users\203>
C:\Users\203>docker start unruffled_brown
unruffled_brown

C:\Users\203>
C:\Users\203>docker inspect unruffled_brown
[
    {
        "Id": "aa103f4d1876a1a23e23964aa1e76089ed7b626ae6b6ec26e508cd388e269340",
        "Created": "2026-07-30T08:11:50.583375269Z",
        "Path": "/bin/bash",
        "Args": [],
        "State": {
            "Status": "running",
            "Running": true,
            "Paused": false,
            "Restarting": false,
            "OOMKilled": false,
            "Dead": false,
            "Pid": 685,
```

Conclusion

In this experiment, Docker was used to understand the fundamentals of containerization. Docker images were downloaded, containers were created and managed, and Linux commands were executed inside an Ubuntu container. The experiment provided practical knowledge of Docker image management, container lifecycle, networking, and basic container operations, helping to understand how Docker provides a consistent and isolated environment for application deployment.